

Week - 09

First Order Logic

Create a knowledge base consisting of FOL statements and prove the given query using Resolution.

Proof by Resolution

Given KB or premises

- John likes all kind of food
- Apple & vegetables are food
- Anything anyone eats & not killed is food
- Anil eats peanuts & still alive
- Harry eats everything that Anil eats
- Anyone who is alive implies not killed
- ~~Any~~ Anyone who is not killed implies alive.

Prove by Resolution that:

John likes peanut

Representation in FOL

- a. $\forall x : \text{food}(x) \rightarrow \text{likes}(\text{John}, x)$
- b. $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
- c. $\forall x \forall y : \text{eats}(x, y) \wedge \neg \text{killed}(x) \rightarrow \text{food}(y)$
- d. $\text{eats}(\text{Anil}, \text{peanuts}) \wedge \text{alive}(\text{Anil})$
- e. $\forall x : \text{eats}(\text{Anil}, x) \rightarrow \text{eats}(\text{Harry}, x)$
- f. $\forall x : \neg \text{killed}(x) \rightarrow \text{alive}(x)$
- g. $\forall x : \text{alive}(x) \rightarrow \neg \text{killed}(x)$
- h. $\text{likes}(\text{John}, \text{peanuts})$

Eliminate implication

- $\forall x \neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
- $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
- $\forall x \forall y \neg [\text{eats}(x, y) \wedge \neg \text{killed}(x)] \vee \text{food}(y)$
- $\text{eats}(\text{Anil}, \text{peanuts}) \wedge \text{alive}(\text{Anil})$
- $\forall x \neg \text{eats}(\text{Anil}, x) \vee \text{eats}(\text{Harry}, x)$
- $\forall x \neg [\neg \text{killed}(x)] \vee \text{olive}(x)$
- $\forall x \neg \text{alive}(x) \neg \text{killed}(x)$
- $\text{likes}(\text{John}, \text{peanuts})$

Move negation ($\neg$) inwards & rewrite

- $\forall x \neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
- $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
- $\forall x \forall y \neg \text{eats}(x, y) \vee \text{killed}(x) \vee \text{food}(y)$
- $\text{eats}(\text{Anil}, \text{peanuts}) \wedge \text{alive}(\text{Anil})$
- $\forall x \neg \text{eats}(\text{Anil}, x) \vee \text{eats}(\text{Harry}, x)$
- $\forall x \text{killed}(x) \vee \text{olive}(x)$
- $\forall x \neg \text{alive}(x) \neg \text{killed}(x)$
- $\text{likes}(\text{John}, \text{peanuts})$

Rename variable or standardize variables

- $\forall x \neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
- $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
- $\forall y \forall z \neg \text{eats}(y, z) \vee \text{killed}(y) \vee \text{food}(z)$
- $\text{eats}(\text{Anil}, \text{peanuts}) \wedge \text{alive}(\text{Anil})$
- $\forall w \neg \text{eats}(\text{Anil}, w) \vee \text{eats}(\text{Harry}, w)$

- f. $\neg g$ killed (g) $\vee$ alive (g)
 g. $\neg k$ $\neg$ alive(k) $\vee$ $\neg$ killed(k)
 h. likes (John, Peanuts)

Drop universe

- a. $\neg$ food (x) $\vee$ likes (John, x)
 b. food (Apple)
 c. food (vegetables)
 d. $\neg$ eats (g, z) $\vee$ killed (y) $\vee$ food (z)
 e. eats (Anil, Peanuts)
 f. alive (Anil)
 g. $\neg$ eats (Anil, w) $\vee$ eats (Harvey, w)
 h. killed (g) $\vee$ alive (g)
 i. $\neg$ alive (k) $\vee$ $\neg$ killed (k)
 j. likes (John, Peanuts)

$\neg$ likes (John, Peanuts) $\neg$ food (x) $\vee$ likes (John, x)
 {Peanuts/x}

$\neg$ food (Peanuts) $\neg$ eats (g, z) $\vee$ killed (y) $\vee$ food (z)
 {Peanuts/z}

$\neg$ eats (y, Peanuts) $\vee$ killed (y) eats (Anil, Peanuts)
 {Anil/y}

killed (Anil) $\neg$ alive (k) $\vee$ $\neg$ killed (k)
 {Anil/k}

$\neg$ alive (Anil) alive (Anil)

$\therefore$ Hence proved.

Week - 10

Algorithm.

1. Input:

- Knowledge Base (KB)

- Query (Q)

2. Convert KB and $\neg Q$ to clausal Form:

- Eliminate implications.

- Move negations inward.

- Standardize variables.

- skolemize (remove $\exists$ quantifiers)

- Drop universal quantifiers

- Convert to CNF

3. Apply Resolution:

- Repeatedly resolve pairs of clauses that contain complementary literals.

- Add new clauses to the KB

- Stop if

- Empty clause (1) is derived $\rightarrow Q$ is true

- No new clauses can be added $\rightarrow Q$ is False

4. Output True / False for Query Q.

QED

27/10/25